Properties of Equality

Jonathan R. Bacolod

Equation

An **equation** is a mathematical statement showing that two expressions are equal.

$$3 + 4 = 7$$

Both sides have the same value.

Equal Sign

The equal sign ($=$) indicates that both sides have the same value.

$$\underbrace{2+5}_{\text{left side}} = \underbrace{7}_{\text{right side}}$$

Properties of Equality

The **properties of equality** are rules that allow us to perform the same operation on both sides of an equation without changing its equality.

Think of a balanced scale: whatever you do to one side, do to the other.

Reflexive Property

Any quantity is equal to itself.

$$a = a$$

Example: $9 = 9$

Symmetric Property

If one quantity equals another, we can reverse their positions.

$$\text{If } a = b, \text{ then } b = a.$$

Example: If $x = 5$, then $5 = x$.

Transitive Property

If two quantities are equal to the same quantity, they are equal to each other.

If $a = b$ and $b = c$, then $a = c$.

Example: If $x = y$ and $y = 4$, then $x = 4$.

Addition Property

Adding the same quantity to both sides preserves equality.

If $a = b$, then $a + c = b + c$.

Example: If $x = 3$, then $x + 2 = 3 + 2$.

Subtraction Property

Subtracting the same quantity from both sides preserves equality.

If $a = b$, then $a - c = b - c$.

Example: If $x = 10$, then $x - 4 = 10 - 4$.

Multiplication Property

Multiplying both sides by the same quantity preserves equality.

If $a = b$, then $ac = bc$.

Example: If $x = 6$, then $2x = 2(6)$.

Division Property

Dividing both sides by the same nonzero quantity preserves equality.

If $a = b$ and $c \neq 0$, then $\frac{a}{c} = \frac{b}{c}$.

Example: If $x = 12$, then $\frac{x}{3} = \frac{12}{3}$.

Substitution Property

A quantity may be replaced by another quantity equal to it.

If $a = b,$ then a may replace b.

Example: If $x = 5$, then $x + 2 = 5 + 2$.

Summary of the Properties

Reflexive	$a = a$
Symmetric	$a = b \Rightarrow b = a$
Transitive	$a = b,\ b = c \Rightarrow a = c$
Addition	$a = b \Rightarrow a + c = b + c$
Subtraction	$a = b \Rightarrow a - c = b - c$
Multiplication	$a = b \Rightarrow ac = bc$
Division	$a = b \Rightarrow \frac{a}{c} = \frac{b}{c}$
Substitution	replace a with b

Memory Aid

Remember **R**-**S**-**T**-**A**-**S**-**M**-**D**-**S**: **R**eflexive, **S**ymmetric, **T**ransitive, **A**ddition, **S**ubtraction, **M**ultiplication, **D**ivision, and **S**ubstitution.

Remember

Whatever operation you perform on one side of an equation, perform the same operation on the other side to maintain equality.

Examples

Example 1

Identify the property illustrated.

$$15 = 15$$

Reflexive Property of Equality

Example 2

Identify the property illustrated.

If $x = 7$, then $x + 3 = 7 + 3$.

Addition Property of Equality

Example 3

Identify the property illustrated.

If $a = b$ and $b = c$, then $a = c$.

Transitive Property of Equality

Example 4

Identify the property illustrated.

If $y = 20$, then $\frac{y}{5} = \frac{20}{5}$.

Division Property of Equality

Practice Exercises

Practice 1

Identify the property of equality illustrated.

$$12 = 12$$

Practice 2

Identify the property of equality illustrated.

If $m = n$, then $n = m$.

Practice 3

Identify the property of equality illustrated.

If $p = q$ and $q = r$, then $p = r$.

Practice 4

Identify the property of equality illustrated.
If $x = 8$, then $x + 5 = 8 + 5$.

Practice 5

Identify the property of equality illustrated.
If $y = 15$, then $y - 4 = 15 - 4$.

Practice 6

Identify the property of equality illustrated.
If $a = 6$, then $3a = 3(6)$.

Practice 7

Identify the property of equality illustrated.

If $b = 20$, then $\frac{b}{4} = \frac{20}{4}$.

Practice 8

Identify the property of equality illustrated.
If $r = s$, then $r + 7 = s + 7$.

Practice 9

Identify the property of equality illustrated.
If $u = v$, then $u - 9 = v - 9$.

Practice 10

Identify the property of equality illustrated.
If $k = 11$, then $k + 2 = 11 + 2$.

Activities

Activity 1

Identify the property of equality illustrated.

$$19 = 19$$

Activity 2

Identify the property of equality illustrated.
If $c = d$, then $d = c$.

Activity 3

Identify the property of equality illustrated.
If $f = g$ and $g = h$, then $f = h$.

Activity 4

Identify the property of equality illustrated.

If $w = 12$, then $w + 6 = 12 + 6$.

Activity 5

Identify the property of equality illustrated.

If $z = 18$, then $z - 7 = 18 - 7$.

Activity 6

Identify the property of equality illustrated.
If $d = 9$, then $5d = 5(9)$.

Activity 7

Identify the property of equality illustrated.

If $c = 24$, then $\frac{c}{6} = \frac{24}{6}$.

Activity 8

Identify the property of equality illustrated.
If $j = k$, then $j + 10 = k + 10$.

Activity 9

Identify the property of equality illustrated.
If $s = t$, then $s - 3 = t - 3$.

Activity 10

Identify the property of equality illustrated.

If $m = 14$, then $m + 5 = 14 + 5$.

Thank you!